

■ Scientific Justification (required for all)

Does circumnuclear dust remember an AGN’s accretion history? An AGN can change its accretion state within years, but the dust surrounding it need not reach a new equilibrium on the same timescale. Changing-look AGN, whose broad optical lines change dramatically [1], offer a direct experiment: after a dated luminosity change, does the infrared spectrum reflect delayed heating of the original dust, or has the dust distribution itself changed? This determines whether infrared emission measures an AGN’s current power or retains evidence of an earlier accretion state. We propose MIRI/MRS spectroscopy of six AGN with contrasting optical and infrared histories to test whether a fixed dust distribution can explain their present spectra.

The physical test. The dust sublimation radius increases approximately as $L^{1/2}$ [2]. After a luminosity decline, surviving dust at the old inner edge cools; newly formed dust closer to the nucleus can instead restore a hot component. During a rise, the sublimation boundary retreats. Reverberation measurements of Mrk 590 suggest that its inner dust distribution recovered within a few years of a decline [3]. Is such restructuring required more generally, or do light-travel delays and a fixed distribution suffice? Infrared echoes establish that dust responds to changing accretion [4], but broadband colours mix temperature, emitting area, extinction and host emission. *The decisive observable is a nuclear dust spectrum whose short-wavelength continuum, warm-dust emission and silicate profiles must all be explained by the same luminosity history and geometry.*

Why MIRI is necessary. Our new SPHEREx spectra already trace the rise into the infrared (Figure 1). They provide the short-wavelength boundary condition, but stop near $5\ \mu\text{m}$ [5]. MRS adds continuous $4.9\text{--}27.9\ \mu\text{m}$ coverage [6]: the hot-to-warm continuum shape, both the 9.7 and $18\ \mu\text{m}$ silicate features, and emission lines and aromatic bands that constrain contamination. Measuring these together is essential: a red continuum alone can be produced by cooler nuclear dust, extinction or star formation. MIRI separates extended host emission where measurable and isolates lines from the continuum; the torus remains unresolved. Archival WISE 12 and $22\ \mu\text{m}$ photometry supplies an earlier baseline, but cannot provide this decomposition. SPHEREx and shorter-wavelength surveys therefore complement, rather than replace, the proposed observations.

Why these targets. The sample combines three anchors with different histories—a faded nucleus, a recovering nucleus and a brightening nucleus—with three comparisons spanning similar redshifts and archival infrared brightness (Table 1). Their multi-epoch optical spectra, including September 2026 observations, establish the changing broad-line and continuum states; ZTF and WISE date the changes between spectra. The targets come from a variability-selected sample [7]. We model each comparison’s measured variability, including slow infrared trends, rather than assume perfect constancy. This design tests whether spectral departures track the direction and elapsed time of a transition, instead of simply reflecting the diversity of AGN dust geometry.

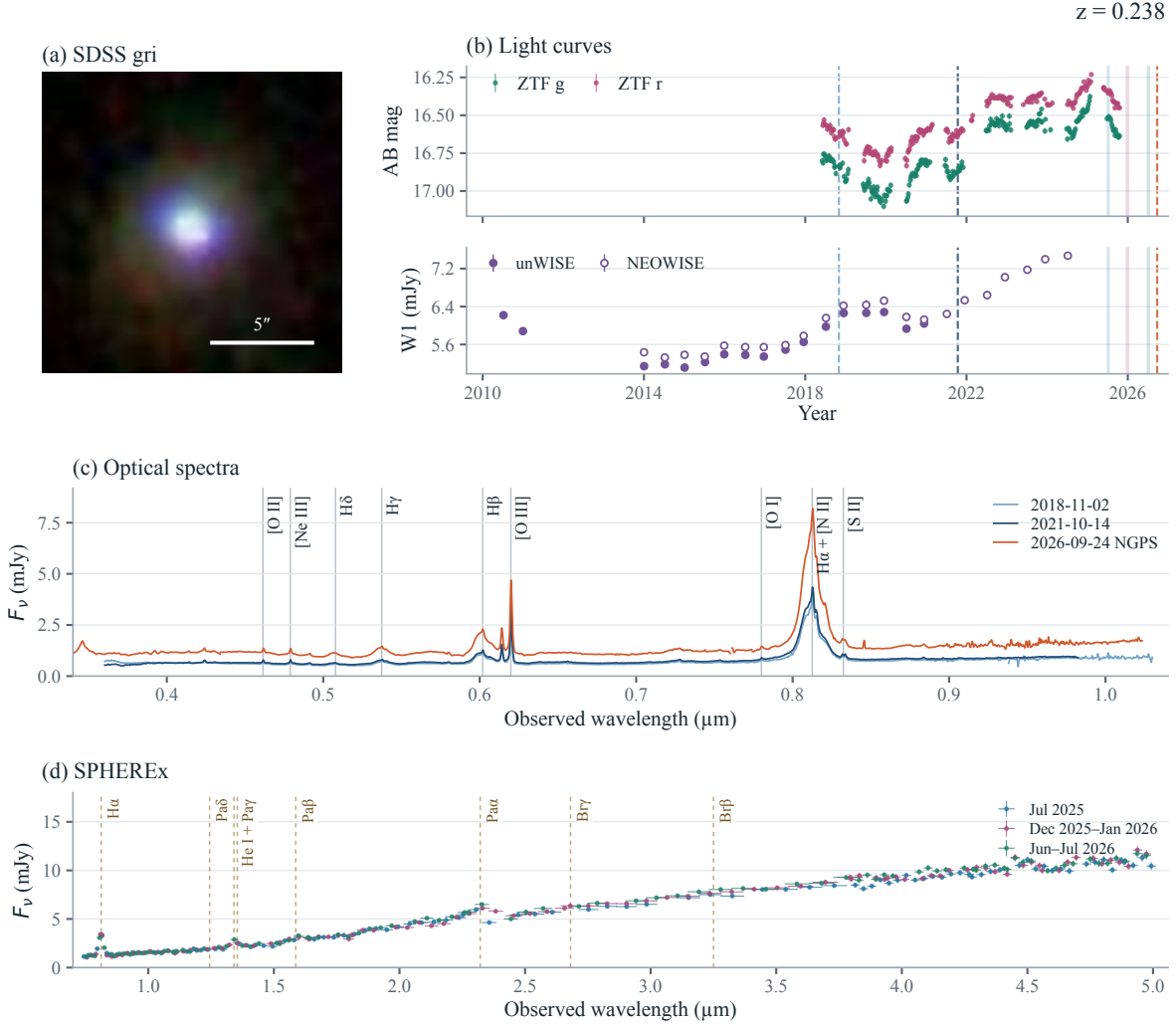


Figure 1: **A3 connects a measured accretion history to the emerging dust spectrum.** The $15''$ SDSS field, light curves, three optical spectra and three SPHEREx intervals show the data in hand. Spectra use observed wavelength and F_v ; line labels mark expected positions. MIRI will connect the rising SPHEREx continuum to the warm dust. The experiment does not require significant variability between SPHEREx epochs.

What JWST adds to this example. For A3 ($z = 0.238$), the two silicate features fall at 12.0 and $22.3 \mu\text{m}$, entirely beyond SPHEREx. MRS will determine whether the rising near-infrared component connects to a nuclear warm-dust spectrum or is diluted by host emission. The 6.2 and $11.3 \mu\text{m}$ aromatic bands and $[\text{Ne II}]$ constrain that mixture; $[\text{Ne V}]$ 14.32 and $[\text{Ne VI}]$ $7.65 \mu\text{m}$ measurements or limits constrain the hard ionizing component [8]. Narrow lines constrain ionization, not the instantaneous luminosity.

■ Description of Observations (required for all GO and Survey)

A common experiment across six histories. All targets have $z < 0.3$ and archival W4 fluxes above 5 mJy. Both silicate peaks lie within MRS; fits account for the limited red wing at the highest redshift. A1 and A2 test the hot component after fading; A3 tests brightening. Comparisons constrain spectral diversity without a large optical transition.

ID	Name	z	History	W3/W4 (mJy)	Si obs. (μm)
A1	J0125+1518	0.112	faded, 2019–21	4.8 / 8.0	10.8 / 20.0
A2	J0053+2232	0.149	faded; recovering	13.6 / 19.3	11.1 / 20.7
A3	J0046+0907	0.238	brightening	23.5 / 74.8	12.0 / 22.3
C1	J0044+1921	0.179	comparison	31.3 / 91.5	11.4 / 21.2
C2	J2349+3029	0.279	slow IR rise	9.2 / 27.6	12.4 / 23.0
C3	J2344+2600	0.163	slow IR decline	6.5 / 18.1	11.3 / 20.9

Table 1: W3/W4 are archival AllWISE fluxes, including host emission, not predicted nuclear MIRI fluxes. Si obs. gives the redshifted silicate peaks.

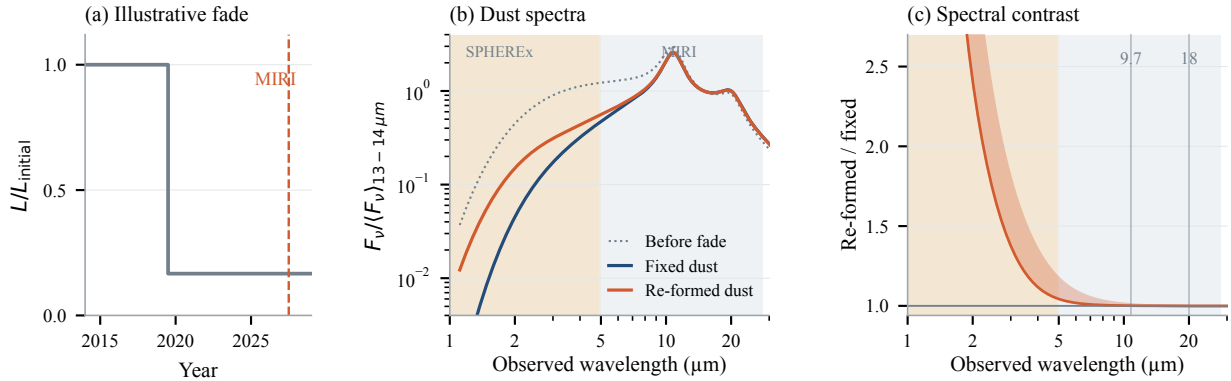


Figure 2: A delayed echo and replenished inner dust need not have the same spectral shape.

An illustrative factor-six fade at A1’s redshift, not a fit to its history. Shells integrate the light-travel response; re-formation adds inner dust. Spectra are normalized at rest 13–14 μm . Shading spans $\rho \propto r^{-1}$ to $r^{-1.5}$; panel (b) shows $r^{-1.5}$. MIRI supplies the warm-dust reference and opacity diagnostics. No instrumental error forecast is implied.

A measurable prediction, with explicit limits. Adding inner dust increases the rest 4.5–5.5 to 13–14 μm continuum ratio by 3–12% in this illustration. Joint fits with the shorter-wavelength spectra constrain that contribution. Departures below the combined model and calibration uncertainty yield limits, not a claimed detection of restructuring.

Observing strategy and precision. Each target receives one MRS visit with all three grating settings, target acquisition and a four-point point-source dither [6,9]. This obtains the diagnostic wavelengths in one visit, avoiding variability between MIRI colours. The planning configuration is FASTR1, 60 groups and one integration per dither, approximately 33 minutes on source. The exposure goals are statistical $S/N \geq 50$ per $R = 100$ continuum bin near rest 5 and 13–14 μm , and ≥ 30 across the 18 μm feature. Broad-band averages then constrain continuum shape, while native-resolution spectra identify contaminating lines. Wavelength-dependent calibration errors and residual fringes are propagated separately; a colour cancels a common normalization error, not every calibration error [10].

The present allocation is approximately 14.1 charged hours, including dedicated skies for A1, A2 and C3. Final ETC/APT calculations must validate this allocation using conservative nuclear spectra and the current channel-4 throughput. Dedicated backgrounds are recommended for MRS [9]; retaining local backgrounds for the other targets requires demonstrably source-free cube regions, otherwise matched sky exposures will be added. The final timing depends on host extent and sky structure. No repeat visits or special timing constraints are requested: the primary measurement is the broad spectrum at a known stage of each history.

From spectra to a falsifiable test. We will fit the MRS cubes with a point source plus extended emission and jointly decompose nuclear continuum, silicates, aromatic bands and lines [8]. Optical spectral decomposition constrains host starlight and nuclear continuum; light curves constrain the intervening evolution. The heating luminosity remains a latent history with uncertainties in host subtraction, spectral shape and unobserved intervals. WISE and SPHEREx are fitted as delayed dust emission, not recycled as an independent driving light curve. Epoch differences and aperture differences are included rather than forcing the data onto one instantaneous spectral energy distribution.

The null hypothesis holds the dust distribution and composition fixed while its emission responds to the inferred illumination with light-travel delays. We span radial distributions, inclination and optical depth using radiative-transfer model families [11,12]. The alternative allows the inner emitting distribution to evolve after the transition. Both must reproduce the continuum, both silicate profiles and the existing infrared history. Obscuration is tested jointly with these constraints: silicate absorption alone neither proves an obscuration-driven transition nor excludes a clumpy torus [11,13]. Comparison objects test whether apparent residuals are common modelling failures. Simulated data with the measured covariance will establish which alternatives the observations can actually distinguish.

The result. A need for additional or depleted hot dust that survives these tests would show that a luminosity echo alone is insufficient and constrain how much the inner emitting distribution changed. If fixed distributions succeed, we will bound the extra hot component and establish which apparent dust changes can be explained by delayed illumination. A single MIRI epoch cannot uniquely recover a grain-formation timescale or dust mass. It supplies the spectral constraints needed to turn the existing time series into a test of dust restructuring.

Supplemental Information

No special timing constraints or coordinated parallel observations are requested.

■ References

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Generative AI disclosure. OpenAI Codex (GPT-6; build identifier not exposed) assisted with review of supplied data and literature, scientific framing, drafting, figure code and document checks on 2026 September 25. An earlier draft recorded assistance from Claude Fable 5.1 (Anthropic) with text, figures and preliminary sensitivity/time estimates. Those estimates are not validated ETC/APT outputs. The proposers are responsible for final verification of the science, references and observing configuration.

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